

# Hydrostatic Equilibrium in a Galaxy Cluster and Turbulent Diffusion of Metals in the Hot Gas

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## 1 Introduction

Galaxy clusters are the largest virialized structures in the Universe (Cimatti et al., 2019), with typical masses of  $10^{14}$ – $10^{15} M_{\odot}$  and sizes of several megaparsecs. They are dominated by dark matter (DM), which accounts for about  $\sim 83\%$  of the total mass, while the baryonic component is mainly in the form of a hot, diffuse plasma known as the intracluster medium (ICM).

The ICM is characterized by high temperatures ( $10^7$ – $10^8$  K) and low densities, emitting strongly in X-rays. It is approximately in hydrostatic equilibrium within the gravitational potential well of the cluster, which is primarily determined by the DM halo.

At the center of many clusters lies the Brightest Cluster Galaxy (BCG), a massive elliptical galaxy that plays a key role in the chemical enrichment of the ICM. Heavy elements, in particular iron, are injected into the gas through Type Ia supernovae (SNIa) and stellar winds. Observations show that the iron abundance typically exhibits a central peak, but extends over scales much larger than the size of the BCG.

This indicates that transport processes are important in redistributing metals throughout the cluster. Turbulent motions, generated by mergers, accretion, and active galactic nucleus (AGN) feedback, are believed to drive diffusion of heavy elements, leading to the relatively smooth abundance profiles observed.

In this work, we investigate two key aspects of galaxy cluster physics. In section 2, we describe the physical model of the distribution of DM, BCG mass, and gas diffusion, and present our simulation method of key quantities. Then we present our results of different models in section 3. Finally, we summarize our conclusions in section 4.

## 2 Model

### 2.1 Hydrostatic Equilibrium and DM Halo Modeling

The gas in the cluster is gravitationally confined by the cluster’s DM potential dominantly and is generally assumed to be in a state of hydrostatic equilibrium (HE). The HE condition reflects a balance between the inward gravitational force and the outward pressure gradient of the gas. In spherical symmetry, this equilibrium is expressed by the equation:

$$\frac{dP(r)}{dr} = -\rho_{\text{gas}}(r) \frac{GM(< r)}{r^2}, \quad (1)$$

where  $P$  is the gas pressure,  $M(< r)$  is the total mass enclosed within a sphere of radius  $r$  centered on the cluster,  $\rho_{\text{gas}}(r)$  is the gas density, and  $G$  is the gravitational constant. Assuming the ICM behaves as an ideal gas,

$$P(r) = \frac{\rho_{\text{gas}}(r) k_B T(r)}{\mu m_p}, \quad (2)$$

where  $k_B$  is the Boltzmann constant,  $T(r)$  the temperature at  $r$ ,  $\mu$  the mean molecular weight, and  $m_p$  the proton mass. Thus, the equation (1) can be rewritten as

$$\frac{d \ln \rho_{\text{gas}}(r)}{dr} + \frac{d \ln T(r)}{dr} = -\frac{\mu m_p}{k_B T(r)} \frac{GM(< r)}{r^2}. \quad (3)$$

For isothermal gas ( $T(r) = \text{constant}$ ), the temperature gradient term  $\frac{d \ln T(r)}{dr}$  can be ignored.

Since the gravitational potential of a cluster is primarily determined by its DM content, we begin by defining the DM distribution. We adopt the well-established Navarro-Frenk-White (NFW) profile:

$$\rho_{\text{DM}}(r) = \frac{\rho_{0,\text{DM}}}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2}, \quad (4)$$

where  $\rho_{0,\text{DM}} = 7.35 \times 10^{-26} \text{ g/cm}^3$  is the central density of DM, and  $r_s = 435.7 \text{ kpc}$  is the scale radius, with virial mass  $M_{\text{vir}} \approx 1.3 \times 10^{15} M_{\odot}$  in our case.

The cumulative DM mass enclosed within a radius  $r$  is obtained by integrating the density distribution:

$$M_{\text{DM}}(r) = \int_0^r 4\pi r'^2 \rho_{\text{DM}}(r') dr', \quad (5)$$

which generate a analytic formula of DM mass

$$M_{\text{DM}}(r) = 4\pi\rho_{0,\text{DM}}r_s^3 \left[ \ln\left(1 + \frac{r}{r_s}\right) - \frac{r}{r+r_s} \right]. \quad (6)$$

For a more realistic scenario of the distribution of baryonic matter in galaxy clusters, we include the BCG component involving its stars. It is located at the center of galaxy cluster, and it is a massive elliptical galaxy whose stellar mass distribution contributes to deepening the central gravitational potential. The stellar distribution of the BCG is modeled using the Hernquist profile, which provides an accurate description of elliptical galaxies. The corresponding mass distribution is given by (Hernquist, 1990)

$$M_{\text{Hern}}(r) = M_{\text{BCG}} \left( \frac{r}{r+r_{\text{Hern}}} \right)^2, \quad (7)$$

where  $M_{\text{BCG}}$  represents the total stellar mass of the central galaxy, and  $r_{\text{Hern}}$  represents the Hernquist scale length, which is determined by the half mass radius  $r_{1/2} = (1 + \sqrt{2})r_{\text{Hern}}$ . The total mass contributing to the gravitational potential is

$$M_{\text{tot}}(r) = M_{\text{DM}}(r) + M_{\text{Hern}}(r). \quad (8)$$

This additional central mass component modifies the gravitational field and leads to a steeper gas density profile in the inner region, indicating that the BCG potential is significant in the central region.

To further refine the physical model, a non-isothermal gas distribution is also considered (see equation (3)). Observational evidence suggests that the temperature of ICM varies with radius. To reflect this behavior, a radially dependent temperature profile is introduced:

$$\frac{T(r)}{T_{\text{ICM}}} = 1.35 \cdot \frac{\left(\frac{x}{0.045}\right)^{1.9} + 0.45}{\left(\frac{x}{0.045}\right)^{1.9} + 1} \cdot \frac{1}{\left[1 + \left(\frac{x}{0.6}\right)^2\right]^{0.45}}, \quad (9)$$

where  $T(r)$  is the gas temperature at radius  $r$ , and  $T_{\text{ICM}} = 8.9 \times 10^7 \text{ K}$  the mean gas temperature of the cluster.  $x = r/r_{500}$  denotes the dimensionless radius where  $r_{500}$  is the characteristic radius that is a half of virial radius  $r_{\text{vir}}/2 = 1.4 \text{ Mpc}$ . This temperature gradient introduces a radial dependence in the thermal pressure, which affects the resulting gas density profile through the hydrostatic equilibrium equation (see equation (3)). Such scenario is closer to realistic model.

## 2.2 Turbulent diffusion of iron in ICM

The metallicity of ICM was enriched gradually because of the production of iron during the evolution processes of massive stars occurring within the BCG. A key question is how metals are transported throughout the ICM. Observationally, the iron distribution appears relatively smooth suggesting an efficient transport mechanism. This distribution is believed to be influenced by mechanism such as turbulence and large scale gas dynamics that stir and spread the metals. To model this process, we use a diffusion equation to describe the evolution of iron density.

The iron abundance  $Z_{\text{Fe}}$  was proportional to the ratio of iron density and gas density

$$Z_{\text{Fe}}(r, t) = 1.4 \frac{\rho_{\text{Fe}}(r, t)}{\rho_{\text{gas}}(r, t)}. \quad (10)$$

To investigate how turbulent diffusion redistributes metals in the ICM, we use the diffusion equation to calculate the evolution of iron density  $\rho_{\text{Fe}}(r, t)$ . Considering turbulent diffusion and source terms  $S_{\text{Fe}}$  which represent iron injection from stellar winds and SNIa in the BCG, the evolution equation of iron density becomes

$$\frac{\partial \rho_{\text{Fe}}(r, t)}{\partial t} = \frac{1}{1.4} \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 D \rho_{\text{gas}}(r, t) \frac{\partial Z_{\text{Fe}}(r, t)}{\partial r} \right) + S_{\text{Fe}}(r, t), \quad (11)$$

here,  $D$  denotes the diffusion coefficient.

To explore the behavior of the model, the diffusion equation is solved numerically under different physical scenarios. We work following a recipe consisting of three steps. First, we study the only diffusion effect. We evolve equation (11) with the source term set to zero  $S_{\text{Fe}} = 0$ . In such scenario, the diffusion coefficient  $D$  is a constant, determined by  $D \sim 0.11 v_t l_t$ , where  $v_t = 260 \text{ km s}^{-1}$  is the characteristic velocity and  $l_t = 15 \text{ kpc}$  the typical spatial scale of the stochastic motion of gas (Rebusco et al., 2005, 2006), and the initial iron abundance adopted the observed iron profile in the Perseus cluster (Rebusco et al., 2005),

$$Z_{\text{Fe}}(r) = 0.3 \times \frac{2.2 + (r/80)^3}{1 + (r/80)^3} \times 1.4 Z_{\odot}, \quad (12)$$

where  $Z_{\odot} = 1.8 \times 10^{-3}$  is the solar metal abundance, and  $r$  is in kpc. Second, neglecting diffusion, we only consider the source term and start with  $Z_{\text{Fe}} = 0$  everywhere. Here, we should consider two kind of stellar feedback from SNIa and stellar winds. The total contribution is determined by

$$S_{\text{Fe}}(r, t) = \rho_{\star}(r) \left[ \frac{\alpha_{\star}(t) Z_{\text{Fe},\star}(r)}{1.4} + \alpha_{\text{Ia}}(t) \cdot \frac{M_{\text{Fe,Ia}}}{M_{\text{Ia}}} \right], \quad (13)$$

where  $\alpha_{\star}$  and  $\alpha_{\text{Ia}}$  are specific mass return rates of stellar wind and SNIa, respectively. Stellar iron content  $Z_{\text{Fe},\star}(r)$  is assumed constant and equal to the solar value  $Z_{\odot}$ , and the  $\rho_{\star}(r)$  denotes the stellar density profile, which is given by the Hernquist model (Hernquist, 1990)

$$\rho_{\star}(r) = \frac{M_{\text{BCG}}}{2\pi} \frac{r_{\text{Hern}}}{r(r + r_{\text{Hern}})^3}. \quad (14)$$

For SNIa, we usually take  $M_{\text{Fe,Ia}} \approx 0.74 M_{\odot}$  and  $M_{\text{Ia}} \approx 1.4 M_{\odot}$ . Generally,  $\alpha_{\star}(t)$  and  $\alpha_{\text{Ia}}(t)$  were large in the past. Defining the supernova unit  $SNu(t) = SNu \left( \frac{t}{t_{\text{now}}} \right)^{-s}$ , where  $SNu$  is the present-day supernova rate expressed per unit B-band luminosity, and  $s$  denotes the slope set of -1.1, the two terms are written as

$$\alpha_{\star}(t) = 4.7 \times 10^{-20} \left( \frac{t}{t_{\text{now}}} \right)^{-1.26}, \quad (15)$$

$$\alpha_{\text{Ia}}(t) = 4.436 \times 10^{-20} \left( \frac{SNu}{M/L_B} \right) \left( \frac{t}{t_{\text{now}}} \right)^{-s}. \quad (16)$$

Here,  $SNu \in [0.05, 0.5]$  since this value is uncertain for elliptical galaxies (BCGs in particular), and  $M/L_B = 7.5$ . The iron injection rate is given by

$$S_{\text{Fe}}(r, t) = \rho_{\star}(r) \left[ \alpha_{\star}(t) \frac{Z_{\text{Fe},\star}}{1.4} + \alpha_{\text{Ia}}(t) z_{\text{Fe,Ia}} \right],$$

where  $z_{\text{Fe,SN}} = M_{\text{Fe,Ia}}/M_{\text{Ia}}$ . In the simulation, the current cosmic time is set as  $t_{\text{now}} \approx 13.7 \text{ Gyr}$ . The normalization of  $\alpha_{\text{Ia}}(t)$  remains uncertain, as it depends on the assumed supernova rate  $SNu$ , and the adopted mass-to-light ratio. Third, combining diffusion and source, we evolve the full equation starting from  $Z_{\text{Fe}} = 0$  everywhere. The density of the gas follows the profile of hydrostatic equilibrium with gradient temperature and BCG contribution. Additionally, we also vary the SNIa unit  $SNu$  and diffusion coefficient  $D$  in a reasonable way to get the best agreement. This represents the most realistic scenario.

If the diffusion of iron affect the distribution of iron metallicity, we can also estimate the timescale for diffusion. The timescale for diffusion is defined as

$$\tau_{\text{diff}} \sim \frac{r_{\text{peak}}^2}{D}, \quad (17)$$

where  $r_{\text{peak}}$  is the width of the iron peak, setting as the radius where the iron metallicity decreases to the half of its maximum. This method is applied to the only diffusion case and diffusion + source case.

## 2.3 Numerical methods

### 2.3.1 Grid construction and discretization

In the simulation, the computational domain is discretized using a one dimensional radial grid extending from an inner boundary of  $r_{\text{min}} = 0.1 \text{ kpc}$  to an outer boundary of  $r_{\text{max}} = 3 \text{ Mpc}$ . We employ a uniform grid

spacing,  $\Delta r = (r_{\max} - r_{\min})/(N_j - 1)$ , where  $N_j = 8000$  represents the total number of radial point. Hence, the position of the  $j$ -th grid interface is defined as  $r_j = r_{\min} + (j-1)\Delta r$ .

To facilitate the numerical solution, we adopt a staggered grid interface. Primary physical quantities (e.g., density, pressure, and temperature) are defined at cell centers,  $r_{j+\frac{1}{2}}$ , while masses and diffusion flux are evaluated at the cell boundaries,  $r_j$ . The cell centers are given by  $r_{j+\frac{1}{2}} = r_j + \Delta r/2$ . Both two set of meshes are used throughout the computation.

Under the assumption of the spherical symmetry, the computational domain is partitioned into a series of concentric shells. The volume  $\Delta V_j$  of the  $j$ -th shell is defined by its radial boundaries:

$$\Delta V_j = \frac{4}{3}\pi(r_j^3 - r_{j-1}^3). \quad (18)$$

Physical quantities (e.g., density, pressure) are defined at the cell centers  $r_{j+1/2}$  and represent the volume-averaged values within each corresponding shell.

### 2.3.2 Mass Integration and Gravitational Field

The enclosed mass is obtained by integrating the DM density and gas density profiles. In the numerical implementation, this integral is approximated by a discrete summation over spherical shells

$$M_{j+1} = M_j + \rho_{j+\frac{1}{2}} \Delta V_j, \quad (19)$$

where the density is evaluated at the cell center.

The BCG is modeled using a Hernquist profile with a total stellar mass  $M_{BCG} = 10^{12} M_\odot$ . The corresponding scale radius is defined as  $r_{\text{hern}} = \frac{r_{1/2}}{1 + \sqrt{2}}$  with  $r_{1/2} = 12$  kpc. The total enclosed mass is then obtained by adding the BCG contribution to the DM component.

Assuming spherical symmetry, the gravitational acceleration at radius  $r$  can be obtained by using the formula after obtaining total mass profile

$$g_j = \frac{GM_{\text{tot},j}}{r_j^2} \quad (20)$$

The gravitational field is then used to solve the hydrostatic equilibrium equation, obtaining gas density which effects the transport of iron. Here, we consider two cases. First, only DM gravity to gas, and second, the masses of DM and BCG contribute to gravity. Under this framework, the density profile of isothermal temperature  $T(r) = T_{\text{ICM}}$  can be calculated via

$$\ln \rho_{\text{gas},j+1/2} = \ln \rho_{\text{gas},j-1/2} - (r_{j+1/2} - r_{j-1/2}) \frac{\mu m_p}{k_B T} g_j, \quad (21)$$

If the temperature distribution follows equation (9), the density profile is

$$\ln \rho_{\text{gas},j+1/2} = \ln \rho_{\text{gas},j-1/2} - (r_{j+1/2} - r_{j-1/2}) \frac{\mu m_p}{k_B T_{j+1/2}} g_j - (\ln T_{j+1/2} - \ln T_{j-1/2}), \quad (22)$$

The central density  $\rho_{\text{gas},3/2}$  is set of  $\rho_0$ . Once the gas density is obtained, the gas mass is computed by integrating the density over spherical shells. The iron density is then derived from the gas density and the iron abundance.

### 2.3.3 Turbulent diffusion to iron contents

In our model, the diffusion coefficient  $D$  is a constant, and the gas density  $\rho_{\text{gas}}(r)$  does not change with time. To solve the diffusion term of the equation (11) numerically, we use the explicit Forward Time Centered Space (FTCS) method following

$$\frac{\rho_{\text{Fe},j+1/2}^{n+1} - \rho_{\text{Fe},j+1/2}^n}{\Delta t} = \frac{1}{1.4} \frac{r_{j+1}^2 D \rho_{j+1}^n (\nabla Z_{\text{Fe}})_{j+1} - r_j^2 D \rho_j^n (\nabla Z_{\text{Fe}})_j}{(r_{j+1}^3 - r_j^3)/3} + S_{\text{Fe}}(r_{j+1/2}, t_n), \quad (23)$$

where  $n$  is the time level,  $t_n$  is the time value at referred time level,  $\rho_j = (\rho_{j-1/2} + \rho_{j+1/2})/2$ , the gradient of iron metallicity is

$$(\nabla Z_{\text{Fe}})_j = \frac{Z_{\text{Fe},j+1/2} - Z_{\text{Fe},j-1/2}}{r_{j+1/2} - r_{j-1/2}}. \quad (24)$$

In addition, for stability, the integration require the timestep

$$\Delta t = C \min \left( \frac{\Delta r^2}{2D} \right), \quad (25)$$

with  $C = 0.4$ . Since the diffusion coefficient  $D$  is a constant in this work,  $\Delta t$  is a fixed value everywhere.

Table 1: Our suite of simulations.

| Model | Description                         | $D$ (cm <sup>2</sup> /s) | $SNu$ |
|-------|-------------------------------------|--------------------------|-------|
| A     | Fiducial Model                      | $1.32 \times 10^{29}$    | 0.15  |
| B     | Large $SNu$                         | $1.32 \times 10^{29}$    | 0.72  |
| C     | Large $D$ and large $SNu$           | $1.06 \times 10^{30}$    | 2.5   |
| D     | Large $D$ and time-varying $SNu(t)$ | $2.4 \times 10^{29}$     | 0.7   |

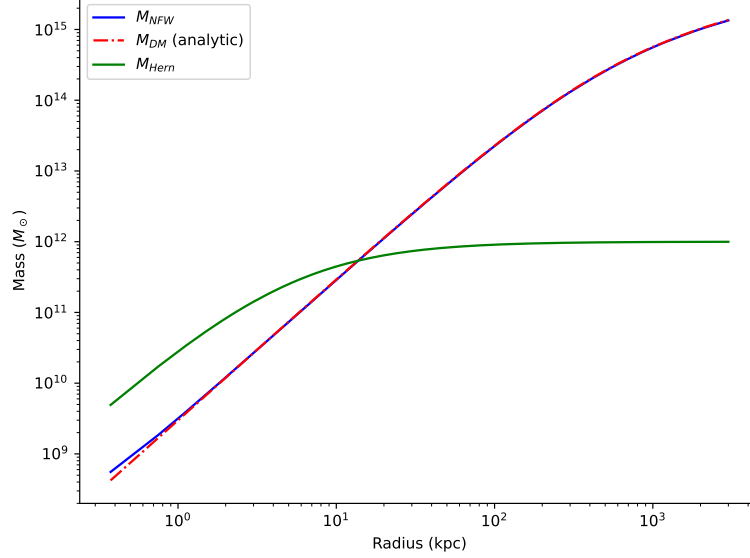


Figure 1: Mass profiles of DM and Hernquist model. The analytic DM model and numerical NFW model are overlapped for compare.

### 2.3.4 Boundary and Initial conditions

Outflow boundary conditions are applied at both inner and outer boundaries. The time integration of equation (23) requires outflow boundary conditions,  $\rho_{\text{Fe},1} = \rho_{\text{Fe},2}$ ,  $\rho_{\text{Fe},j\text{max}} = \rho_{\text{Fe},j\text{max}}$ ,  $Z_{\text{Fe},1} = Z_{\text{Fe},2}$ ,  $Z_{\text{Fe},j\text{max}} = Z_{\text{Fe},j\text{max}}$ .

To conduct the simulation about the turbulent diffusion of Fe in the ICM, we set a set of parameters (model A) following to study the temporal evolution of metallicity, which is simulated over timescales of 1, 2, 5 Gyr to analyze how iron distribution varies over cosmic time within the cluster environment. This simulation is based on prior recipe, with the start time is 8.7 Gyr. Except the parameters of fiducial model, we also vary some parameters and predict its variation at longer cosmic time (such as 100 Gyr) in source + diffusion case for a global understanding of the realistic picture. These variations, along with the additional model runs discussed in Section 3.2.2, are listed in Table 1.

## 3 Results

### 3.1 Part I: hydrostatic equilibrium in a galaxy cluster

To obtain the ICM gas density distribution, we integrate the density distribution using the NFW profile (equation (4)) for the mass of DM, since the ICM gas is attracted by massive matters. Additionally, we also have the mass of BCG (equation (7)). The mass profiles of such two components are plotted in Figure 1. For comparing the numerical solution of DM mass and its analytic one, we overlap the analytic solution (equation (6)). The numerical result is almost same with analytic mass profile. With the mass distribution of DM and BCG, we try to solve the hydrostatic equilibrium equation (equation (1)) to acquire the gas density profile. The results are shown in Figure 2. Firstly, we only consider the gravity of DM for isothermal gas, showing a relatively smooth radial profile, and the density is lower in the central region. We also compare the numerical and analytic solutions of gas density in such scenario, shown in Figure 3. The analytic gas density is read

$$\rho_{\text{gas}}(r) = \rho_0 e^{-27b/2} \left(1 + \frac{r}{r_s}\right)^{27b/(2r/r_s)}, \quad (26)$$

$$b = \frac{8\pi G \mu m_p \rho_{0,DM} r_s^2}{27k_B T}. \quad (27)$$

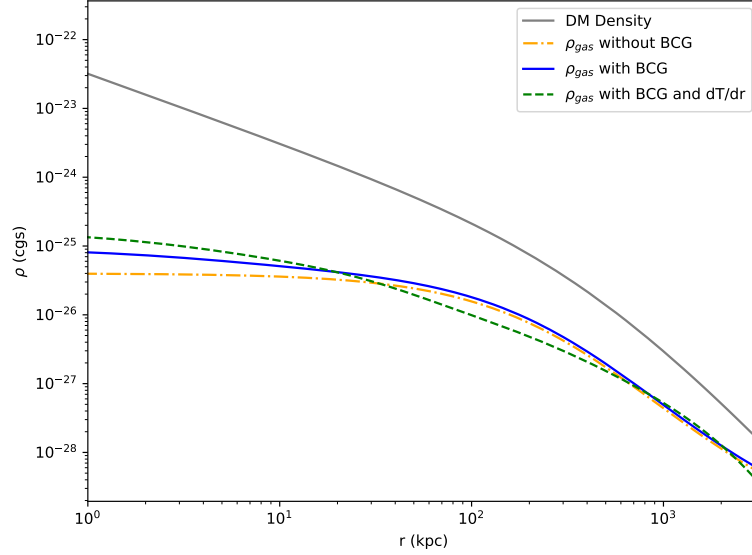


Figure 2: Density profiles of DM and gas in three diverse situation, isothermal gas only attracted by DM, isothermal gas attracted by DM and BCG, and non-isothermal gas attracted by DM and BCG.

where  $\rho_0$  is the central density of gas. The two solutions are precisely superimposed on each other. Second, we add the contribution of BCG to gravity, revealing that the density is higher in central region while in outer region the density distribution is similar to prior one. This is due to the fact that the BCG mass is concentrated in limited radius, providing a stronger gravitational field with respect to the model where we only consider DM, thus, higher gas density are needed to provide pressure against gravity, forming Hydrostatic equilibrium. The gas density of central region are influenced significantly. Third, we consider the temperature distribution characterized by equation (9), which is illustrated in Figure 4. This situation exhibits the highest density in the central region within 10 kpc, but the density is lowest between 20 to 600 kpc, and in the outermost region, the density is close to isothermal cases.

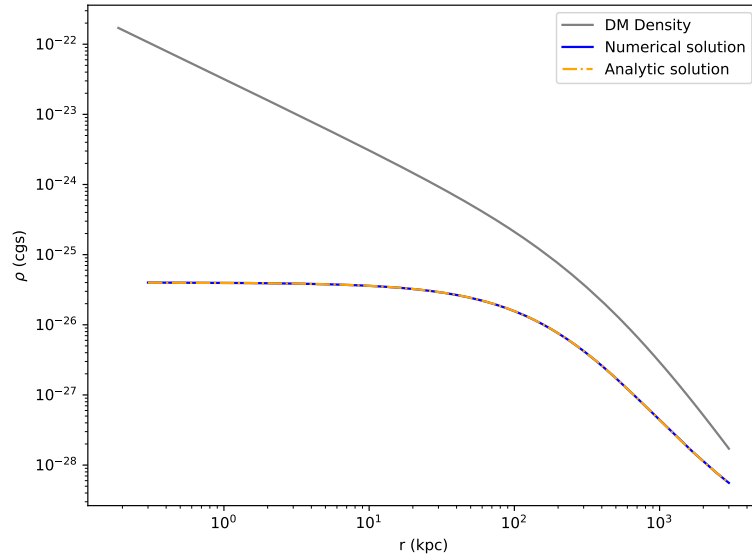


Figure 3: Density profiles of DM and gas in three diverse situation, isothermal gas only attracted by DM, isothermal gas attracted by DM and BCG, and non-isothermal gas attracted by DM and BCG.

For comparison with [Rebusco et al. \(2005\)](#), we plot our results of density profile and gas temperature distribution in Figure 4, and overlap the result from [Rebusco et al. \(2005\)](#).

Additionally, we also check the baryonic fraction  $f_b$ , which is defined as the ratio of baryonic mass and total mass, namely

$$f_b(r) = \frac{M_{\text{Hern}}(r) + M_{\text{gas}}(r)}{M_{\text{DM}} + M_{\text{Hern}}(r) + M_{\text{gas}}(r)}. \quad (28)$$

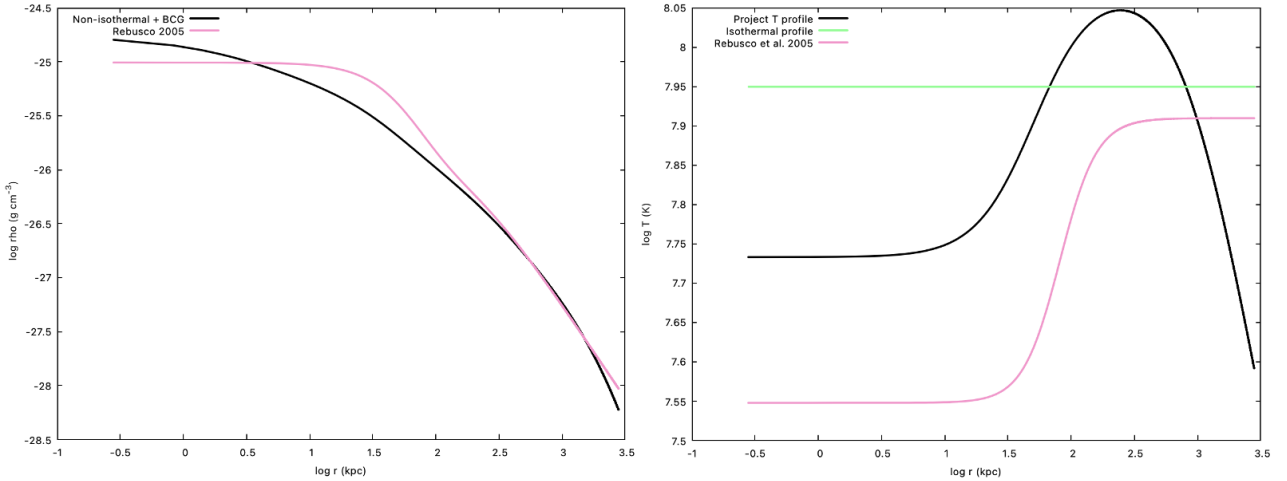


Figure 4: The non-isothermal density profile with gravity of DM and BCG and temperature profile, and comparison with Rebusco et al. (2005), in which the temperature profile is equation (2) and density profile is equation (1) with  $\rho_{\text{gas}} = \mu_e m_p n_e$  in Rebusco et al. (2005).

For our three different situations, the mass distributions of diverse components and the  $f_b$  profiles are shown Figure 5. The  $f_b$  of the three situations in the outermost edge are 0.151, 0.166, and 0.155, with  $\rho_0 = 4.0 \times 10^{-26} \text{ g cm}^{-3}$ ,  $9.0 \times 10^{-26} \text{ g cm}^{-3}$ , and  $1.6 \times 10^{-25} \text{ g cm}^{-3}$ , respectively. Generally, the  $f_b$  exhibits a gradual decline from the cluster center towards the outskirts, and these values are close to the cosmological value  $f_b = 0.169$  at large radii (Cimatti et al., 2019).

### 3.2 Part II: turbulent diffusion of Iron in the ICM

We present the results of our simulations for the evolution of iron abundance in the ICM. We consider three physical scenarios: diffusion only, source only, and the combination of diffusion and source. We first analyze the fiducial model, and then explore how the iron abundance profile varies with different parameters.

#### 3.2.1 Fiducial model

We present in this subsection the main results of the fiducial model. We summarize the evolution of the metallicity in three different scenarios, (i.e. only diffusion, only source, and diffusion + source), and estimate the width of the iron peak as well as the diffusion timescale.

The results of the first step in our recipe which considers diffusion only is illustrated in Figure 6, showing the evolution of the iron abundance profile at different times. Initially, the iron was concentrated in the central region within about 20 kpc. However, the iron abundance decreases with time between 20 and 200 kpc, indicating that iron is transported outward due to turbulent diffusion. This behavior is further supported by the decrease in iron mass within 100 kpc and the boardening of the iron peak. These result are summarized in Table 2, accompanied by the iron mass inside peak, the index of peak, and the diffusion timescale. The total iron mass  $M_{\text{Fe,tot}}$  remains conserved throughout the simulation.

Additionally, we simulate the evolution of iron abundance produced by stellar winds and SNIa, without including diffusion. For simplicity, the temporal variations of  $\alpha_*$  and  $\alpha_{\text{Ia}}$  are neglected and treated as constants. Under this assumption, equation (13) can be simplified as

$$S_{\text{Fe}}(r) \simeq (6 \times 10^{-23} + 4.7 \times 10^{-22}) \rho_*(r), \quad (29)$$

since  $SNu(t) = SNu = 0.15$ . The result of source case of model A is presented in Figure 7. At the same radius, the iron abundance increase with time. The iron abundance also drops dramatically within 1.5 kpc. The theoretical iron mass can be estimated by

$$M_{\text{Fe}}^{\text{theo}} = 5 \times 10^{-22} M_{\text{BCG}} t. \quad (30)$$

Our simulated iron masses  $M_{\text{Fe,tot}}$  listed in Table 2 are very close to the theoretical values,  $1.67 \times 10^7 M_{\odot}$ ,  $3.34 \times 10^7 M_{\odot}$ ,  $8.33 \times 10^7 M_{\odot}$ , for 1, 2, 5 Gyr, respectively.

Finally, the simulation takes diffusion and source into account. In such scenario, we try to simulate longer duration, up to 100 Gyr. This plot is demonstrated in the left panel of Figure 8, showing the effects of both

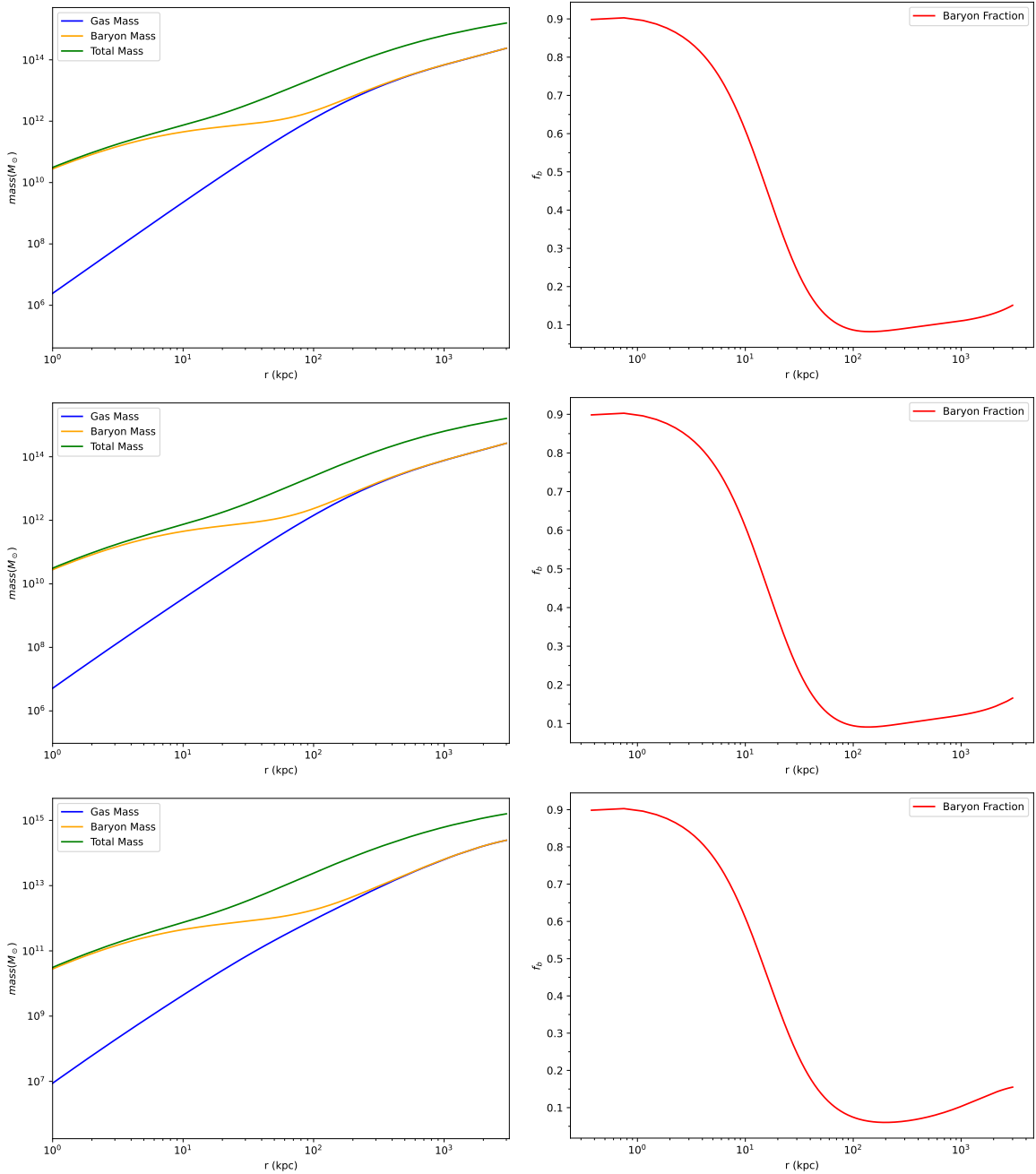


Figure 5: The mass distributions and  $f_b$  profiles in three diverse situations. The top panel shows the isothermal gas with only DM gravity case, the middle panel show the isothermal gas with DM and BCG gravities case, and the bottom panel exhibit the non-isothermal gas with DM and BCG gravities case. The left panel shows the mass distributions, and the right panel indicates the  $f_b$  profiles.

diffusion and source. With evolution, the iron is diffused to outer region, and at the same position, the iron abundance increase. The profile is similar to the observed one, which is flat initially, following a decreasing trend, and become flat around  $Z_{\text{Fe}} = 0$  finally. The iron mass profiles at diverse cosmic times are also present in the right panel of Figure 8, showing the increase of iron mass. The theoretical iron masses at different time are also estimated by equation (30), so that the theoretical iron masses at 10, 24, 100 Gyr are  $1.67 \times 10^8 M_{\odot}$ ,  $4.01 \times 10^8 M_{\odot}$ , and  $1.67 \times 10^9 M_{\odot}$ , respectively.

### 3.2.2 Parameter dependence

We also try to change parameters to find a good fit of observed iron profile. We focus on the profile at 5 Gyr with varying parameters. The result of models B and C are present in Figure 9. Model B reveals that the metal abundance will be enhanced significantly with large  $SNu$ , since stellar processes provide more ingredient.

### Iron abundance with only diffusion

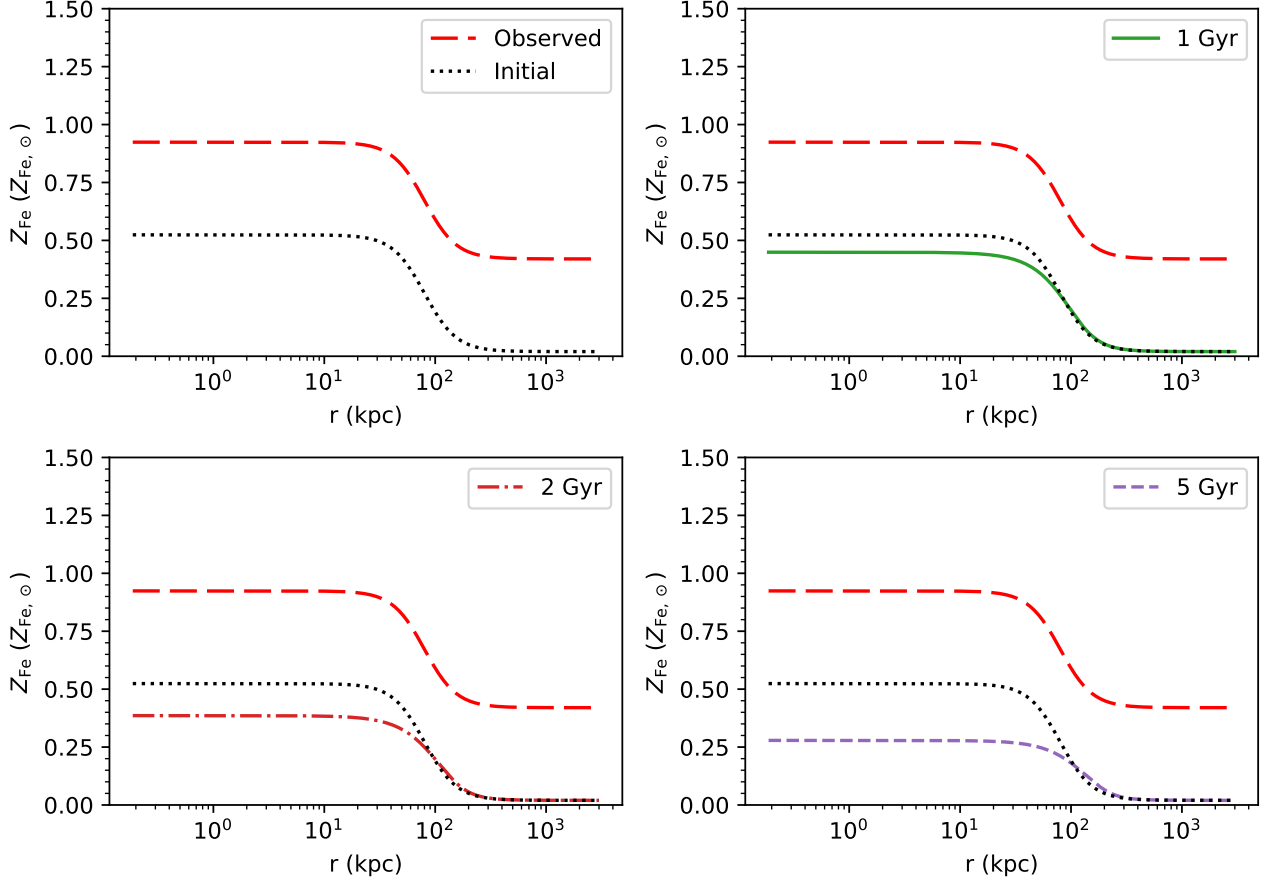


Figure 6: Time evolution of the iron abundance profiles in no source, only diffusion case with fiducial parameters (model A). The initial profile is based on the observation result shown in equation (12). The iron abundance profiles of 0, 1, 2, 5 Gyr are shown in the panels from upper left to bottom right. The initial (0 Gyr) profile is the observed profile subtracting the background abundance  $Z_{\text{Fe,out}} = 0.4Z_{\text{Fe},\odot}$ .

Table 2: The simulated result. The ‘-’ denotes NAN because only source case do not have diffusion timescale  $\tau_{\text{diff}}$ .

| Model                | Time (Gyr) | $M_{\text{Fe,tot}} (M_{\odot})$ | $M_{\text{Fe,100kpc}} (M_{\odot})$ | $j_{\text{peak}}$ | $r_{\text{peak}} (\text{kpc})$ | $M_{\text{Fe,peak}} (M_{\odot})$ | $\tau_{\text{diff}} (\text{Gyr})$ |
|----------------------|------------|---------------------------------|------------------------------------|-------------------|--------------------------------|----------------------------------|-----------------------------------|
| A (only diffusion)   | 0          | $7.13 \times 10^9$              | $3.83 \times 10^8$                 | 220               | 82.32                          | $2.83 \times 10^8$               | 15.47                             |
|                      | 1          | $7.13 \times 10^9$              | $3.49 \times 10^8$                 | 245               | 91.51                          | $3.08 \times 10^8$               | 19.12                             |
|                      | 2          | $7.13 \times 10^9$              | $3.21 \times 10^8$                 | 272               | 101.64                         | $3.28 \times 10^8$               | 23.58                             |
|                      | 5          | $7.13 \times 10^9$              | $2.60 \times 10^8$                 | 343               | 128.27                         | $3.80 \times 10^8$               | 37.56                             |
| A (only source)      | 1          | $1.67 \times 10^7$              | $1.52 \times 10^7$                 | 4                 | 1.13                           | $5.33 \times 10^5$               | -                                 |
|                      | 2          | $3.33 \times 10^7$              | $3.03 \times 10^7$                 | 4                 | 1.13                           | $1.07 \times 10^6$               | -                                 |
|                      | 5          | $8.32 \times 10^7$              | $7.58 \times 10^7$                 | 4                 | 1.13                           | $2.66 \times 10^6$               | -                                 |
| A (diffusion+source) | 1          | $1.66 \times 10^7$              | $1.50 \times 10^7$                 | 25                | 9.00                           | $9.22 \times 10^5$               | 0.18                              |
|                      | 2          | $3.33 \times 10^7$              | $2.94 \times 10^7$                 | 30                | 10.88                          | $1.66 \times 10^6$               | 0.27                              |
|                      | 5          | $8.32 \times 10^7$              | $6.44 \times 10^7$                 | 37                | 13.50                          | $3.29 \times 10^6$               | 0.42                              |
|                      | 10         | $1.66 \times 10^8$              | $1.04 \times 10^8$                 | 45                | 16.50                          | $5.92 \times 10^6$               | 0.62                              |
|                      | 24         | $4.00 \times 10^8$              | $1.65 \times 10^8$                 | 58                | 21.38                          | $1.24 \times 10^7$               | 1.04                              |
|                      | 100        | $1.66 \times 10^9$              | $2.86 \times 10^8$                 | 89                | 33.00                          | $4.11 \times 10^7$               | 2.49                              |
| B                    | 5          | $3.63 \times 10^8$              | $2.81 \times 10^8$                 | 37                | 13.50                          | $1.44 \times 10^7$               | 0.42                              |
| C                    | 5          | $1.24 \times 10^9$              | $3.83 \times 10^8$                 | 67                | 24.75                          | $3.49 \times 10^7$               | 1.39                              |
| D                    | 8          | $6.38 \times 10^8$              | $3.15 \times 10^8$                 | 59                | 21.75                          | $3.49 \times 10^7$               | 1.08                              |

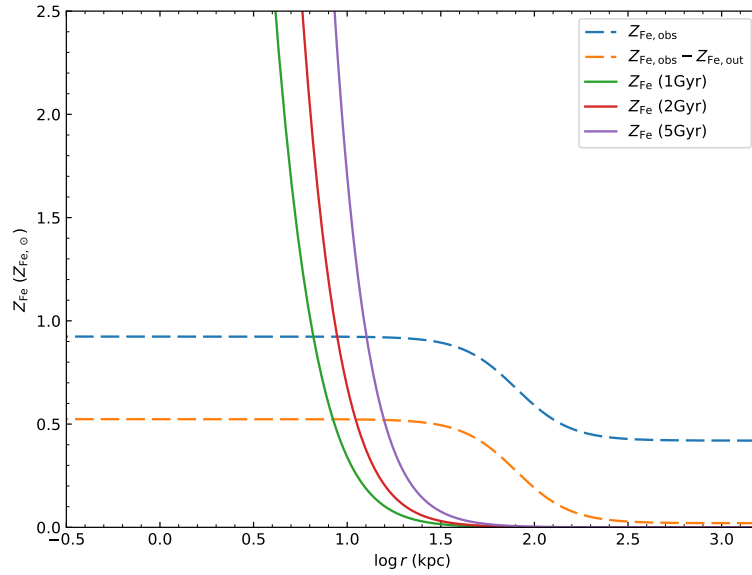


Figure 7: Time evolution of the iron abundance profiles in no diffusion, only source case with fiducial parameters (model A). The observed profile is based on the observation result shown in equation (12).

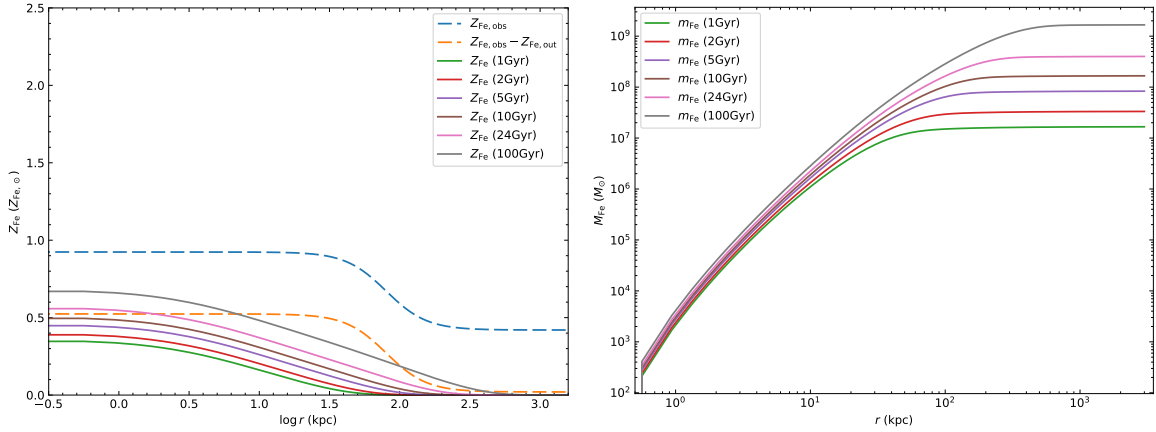


Figure 8: The left panel shows the time evolution of the iron abundance profiles in case combined diffusion and source with fiducial parameters (model A). The observed profile is based on the observation result shown in equation (12). The right panel exhibits the iron mass profiles at different cosmic times, from 1 - 100 Gyr.

Model C shows that if the turbulent diffusion is more efficient, more iron will be transport to outer region. The result of model C is close to the observed profile with the radius  $r > 1.7$  kpc.

If we take the temporal varying  $SNu(t)$  in to account with  $SNu = 0.7$ , which are regarded as model D, we find that at 8 Gyr, the iron content profile also fit the observed profile well. However,

## 4 Conclusion

In this work, we have systematically investigated the gas density distribution and the chemical enrichment history of the ICM in galaxy clusters. By considering various physical components including the NFW DM halo, the Hernquist BCG stellar mass, and radial temperature gradients, we solved the hydrostatic equilibrium equation and simulated the long-term evolution of iron abundance under the influence of turbulent diffusion and stellar sources. Our main findings are summarized as follows.

1. The inclusion of the BCG significantly modifies the central gravitational potential, leading to a steeper gas density profile within the inner 100 kpc compared to a DM-only model. Furthermore, the non-isothermal model (gradient temperature) produces the highest central gas density, which better aligns with the observed cool-core structures. The resulting baryonic fractions ( $f_b \approx 0.15 - 0.17$  at large radii) are consistent with the cosmic baryon fraction, validating our HE framework.

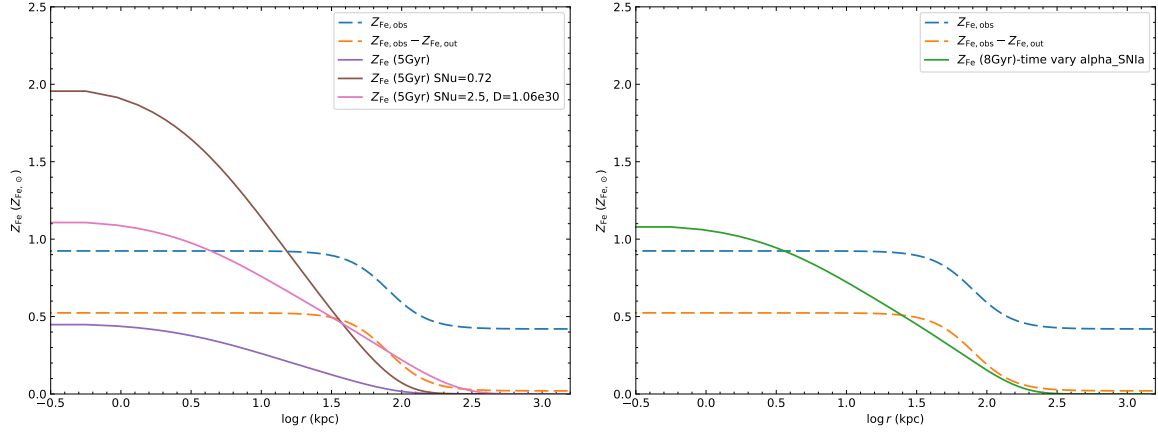


Figure 9: The left panel shows the iron abundance profiles of diffusion + source scenario at 5 Gyr, with parameters of models B and C, while the right panel shows the iron abundance content profiles of diffusion + source scenario at 8 Gyr, with parameters of model D. The observed profile is based on the observation result shown in equation (12).

2. Our diffusion-only simulations demonstrate that turbulent mixing is highly efficient at smoothing out the initial sharp metallicity peaks. The source-only models produce unrealistic, extremely sharp iron distributions restricted to the vicinity of the BCG. Only by combining the iron injection from SNIa and stellar winds with an appropriate diffusion coefficient ( $D \sim 10^{29} \text{ cm}^2/\text{s}$ ), we can reproduce the smooth, extended metallicity profiles observed in systems like the Perseus cluster.

3. Our parameter study (Models B-D) highlights the degeneracy between the supernova rate ( $SNu$ ) and the diffusion coefficient ( $D$ ). A higher  $SNu$  can offset for stronger diffusion to maintain the observed iron peak, suggesting that the enrichment history of clusters is intimately linked to the mechanical stirring of the gas.

Our results show that the ICM is not a static reservoir but a dynamic environment where metals serve as excellent tracers of past energetic activities, including AGN feedback and cluster mergers. Future work incorporating spatially dependent diffusion coefficients and time-evolving BCG star formation rates will provide deeper insights into the co-evolution of galaxies and their host clusters.

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